

CÉSAR AUGUSTO DOS SANTOS MAFRA

Senior Backend Engineer · Python · AI Integration · PostgreSQL

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PROFESSIONAL SUMMARY

Senior Software Engineer with 5+ years of hands-on experience designing Python backends and modernizing critical platforms in government and judicial environments. Specialized in REST API architecture with Django/Flask, optimized PostgreSQL data modeling, and production AI/ML integration (WhisperX/CUDA, LLMs via Gemini/GPT).

Recently served as technical architect for two AI initiatives at the Court of Justice of Acre, including an offline GPU-based judicial transcription pipeline and an LLM-assisted public procurement system. Coordinated a mixed engineering and L1/L2 support team during an interim leadership period.

Holds active Oracle Cloud Infrastructure certifications (Foundations + AI Foundations 2025), with continuing education in Containers/Kubernetes (LINUXtips PICK) and a postgraduate program in Cybersecurity – Red Team (FIAP).

PROFESSIONAL EXPERIENCE

Senior Software Engineer

SONDA Brazil — Court of Justice of Acre (TJAC) | Oct 2025 — May 2026

Python · Flask · Django · PostgreSQL · WhisperX · CUDA · LLMs · Docker · GitLab CI/CD · Keycloak

- ADA/TranscriptJAC: architected and implemented a judicial transcription and speaker diarization pipeline (WhisperX + Wav2Vec2 + Pyannote) running on NVIDIA A2000 GPU, deployed fully offline within the Court's segregated network. Reduced manual transcription from hours to minutes.
- LÚCIA project (AI-assisted public procurement): designed the full Flask backend — PostgreSQL data modeling for DOD/ETP/TR documents, CRUD and dynamic filtering, document versioning, runtime-switchable LLM backends (Gemini and GPT), Keycloak + JWT authentication, Swagger/OpenAPI documentation, and pytest test coverage.
- Vera System (judicial case management): modernized core modules — audit dashboard with request log visualization and CSV export, phonetic name search, advanced filtering with Flatpickr + DataTables, and optimized PostgreSQL views for SAJ/eProc integration (queries that previously took minutes now execute in seconds).
- ML infrastructure: configured and stabilized the Linux/WSL2 + NVIDIA Drivers + CUDA + Conda + PyTorch stack for offline ASR model execution, overcoming network restrictions through dependency pre-packaging.
- DevOps & Deployment: maintained GitLab CI/CD pipelines, Nginx and PM2 configurations, and backend deployments to the Court's production environment.
- Interim technical leadership: acted as technical coordinator for 20 days in December 2025, leading a mixed dev + L1/L2 support team — SLA tracking, task distribution, and operational KPI reporting.

Software Developer

State of Acre Government — Iteracre (Land Institute) | Mar 2024 — Oct 2024

PHP · CakePHP · PostgreSQL · Docker · GitLab

- Designed and built the State Land Registration System, automating land demarcation and title issuance for urban and rural areas.
- Implemented containerized environments with Docker, ensuring development/production parity.
- Established versioning and CI workflows on GitLab for the engineering team.

Data Analyst · Web Developer

XLand Holding (Remote) | Sep 2022 — Dec 2022

Python · PHP · Laravel · SQL

- Developed web modules in Laravel and provided specialized IT infrastructure support.
- Managed complex SQL databases with a focus on data integrity and BI reporting.

Software Developer & IT Division Chief

State of Acre Government — SEASDHM | Feb 2019 — Aug 2022

- Production e-learning platform: built a Python-based Distance Learning platform integrated with Moodle (custom plugins + integrations with internal systems), used by dozens of government employees.
- Digitized internal administrative workflows with Python + PostgreSQL, reducing manual rework in service requests and reporting.
- Led the IT team at the Secretariat of Human Rights and Women's Policies: support, governance, infrastructure, and technical demand management.

TECHNICAL STACK

- Backend & Languages: Python (Django, Flask, FastAPI), Java, advanced SQL, JavaScript/TypeScript
- Data: PostgreSQL (views, optimized queries, indexing), Pandas, NumPy, ETL pipelines
- Applied AI / ML: WhisperX, Faster-Whisper, Pyannote, CUDA/cuDNN, LLM integration (Gemini, GPT), prompt engineering
- Cloud & DevOps: Docker, GitLab CI/CD, Nginx, PM2, Linux, AWS (EC2, S3), Oracle Cloud Infrastructure
- Security: Keycloak, JWT, OWASP principles, Red Team fundamentals (FIAP postgraduate)
- Other: Git, Swagger/OpenAPI, pytest, REST APIs, legacy system integration

PERSONAL PROJECTS

- DevOps Roadmap Tracker — Web app (HTML/JS) with push notifications and localStorage-backed interactive checklists. [dev-ops-roadmap-tracker.vercel.app](#)
- NFLStatsLab — Data Science project analyzing 20 years of NFL data (2003-2023) with Pandas, NumPy, and Seaborn.

EDUCATION

- Postgraduate in Cybersecurity — Red Team · FIAP · In progress (started Aug 2025)
- B.S. in Information Systems · Estácio Unimeta · Completed Jul 2025

CERTIFICATIONS

Certification	Issuer	Date
OCI 2025 Foundations Associate	Oracle	Sep 2025
OCI 2025 AI Foundations Associate	Oracle	Sep 2025

Both valid through Sep 2027 · Oracle Next Education program

CONTINUING EDUCATION

- PICK — Intensive Program in Containers & Kubernetes · LINUXtips · In progress
- Linux for Cloud Native – LINUXTips · In progress
- Full Stack Python Developer · EBAC · 2023/2024

LANGUAGES

Portuguese (Native) · Spanish (Advanced) · English (Intermediate — fluent technical reading) · German (A1)